Robust Prediction of Alcoholism from EEG Signals Using Auto-Encoder

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***Abstract*—The diagnosis of alcoholism (substance abuse) is crucial due to its impact on individuals, society, and healthcare systems worldwide. Electroencephalography (EEG) signals have been shown to contain valuable information about the effects of alcoholism on the brain, and can potentially be used as a non-invasive tool for early detection and monitoring. In this study, we proposed an auto-encoder model for predicting alcoholism from EEG signals, which achieved robust and accurate performance even in the presence of noise and variability in the data. We evaluated the model using multiple classification algorithms, including Logistic Regression, Random Forest Classifier, AdaBoost Classifier, and AutoEncoders. The auto-encoder model (consisting of Bi-LSTMs and ReLU, Sigmoid activation for Dense and output layers respectively) outperformed the other algorithms, achieving an accuracy of 65.5% on the test set, while Logistic Regression achieved an accuracy of 55%, Random Forest Classifier achieved 58%, and AdaBoost Classifier achieved 58.4%. These results demonstrate the effectiveness of auto-encoder models for biomedical signal processing and provide a promising avenue for future research in this area, including the investigation of more complex architectures and feature engineering techniques, and the use of larger and more diverse datasets for training and evaluation. This research contributes to the fields of biomedical signal processing, neurophysiology, and neuroimaging, offering insights into brain activity related to alcoholism and highlighting the importance of non-invasive diagnosis and early detection in tackling alcoholism disorders and their societal impact.**

Keywords*—Electroencephalography (EEG) signals, Auto-encoder model, Bi-directional Long-Short Term Memory (Bi-LSTM), Rectified Linear Unit (ReLU), Dense layer, Logistic Regression, Random Forest Classifier, AdaBoost Classifier, Neurophysiology

I. Introduction

Alcoholism is a global health issue with significant societal and healthcare ramifications. Detecting alcoholism early can help prevent long-term physical and mental consequences. Electroencephalography (EEG) signals, which reflect brain activity, have shown potential for non-invasive alcoholism detection. Recent advancements in AI enable effective feature extraction from biomedical signals, including EEG. In this study, we propose utilizing an auto-encoder, a deep learning model, to predict alcoholism from EEG signals. Auto-encoder models have demonstrated success in various domains such as image and speech processing, and we hypothesize that they can be applied to EEG signals to accura--tely classify alcoholism.

The auto-encoder model utilizes the compressed latent features obtained from the input EEG signals to capture patterns and extract relevant information for alcoholism detection, even in the presence of noise and variability. The model surpasses various classification algorithms, employed in the paper for comparison, such as Logistic Regression, Random Forest Classifier, and AdaBoost Classifier which demonstrates its robustness and accuracy. Our study reveals that the auto-encoder model achieves a predictability rate of 65.5%, outperforming other algorithms. These findings open avenues for future research, including exploring more complex architectures, feature engineering techniques, and employing larger and diverse datasets for evaluation and training purposes.

The remaining parts of this paper is structured in this fashion: Section II gives a concise summary of prior research in EEG-based alcoholism detection. Section III describes the methodology used in this study, including the dataset, preprocessing, and the auto-encoder model architecture. Section IV presents the experimental results and performance evaluation of the proposed model. Our study's implications and potential future research directions in this area are discussed in Section V. The research paper is concluded in Section VI, which also lists this research's contribution.

II. Literature Review

To evaluate the most substantial developments in the subject/field, a literature study is conducted. While some of them have used simple techniques, others have used more complex ones like deep learning and neural networks. These works teach us the basic and complex concepts of artificial intelligence and machine learning, which cover topics like natural language processing, assembly, transfer learning, etc.

The study [1] demonstrates the effectiveness of unsupervised transfer learning using MLP and CNN techniques between the electroencephalographic (brainwave) classification and electromyographic (muscular wave) domains. In order to perform this, 3-class mental state EEG dataset and EMG data from a 4-class gesture classification experiment utilising the Myo Armband and Muse EEG Headband, respectively, are gathered. Additionally, signals from the forearm muscles and the brain are examined. An evolutionary search method called hyperheuristic multi-objective search is used to find the best network hyperparameters. Then, utilising this deep neural network's

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optimised design, they categorise both EMG and EEG data, with results of 84.76% and 62.37% accuracy, respectively. When pre-trained weights from the EMG classification model are utilised for initial distribution rather than random weight initialization, 93.82%(+29.95) accuracy for EEG classification is achieved. When EEG pre-trained weights are used for the initial weight distribution for EMG, 85.12% (+0.36) accuracy is reached.

Gaowei Xu et al. [2] suggest a deep transfer convolutional neural network (CNN) framework based on VGG-16 for the categorization of EEG data. The proposed framework comprises an ImageNet-trained VGG-16 CNN model and a target CNN model that is fundamentally the same as VGG-16 except for the softmax output layer. The target CNN model, which is used to categorise MI EEG data, directly inherits the parameters from the pre-trained VGG-16 CNN model. The target MI dataset then modifies the values in later layers while the front-layer parameters of the target model are set to zero. The target dataset consists of time-frequency spectrum pictures of EEG signals.

In their paper [3], the scientists first denoised the input data using a multilayer discrete wavelet transform (DWT), and then they extracted features using CNNs and Bi-LSTMs. For the CNN-Bi-LSTM network to classify alcohol EEG signals, the denoised data were introduced. With an average classification accuracy of 99.32%, an accuracy of 99.01%, and a recall rate of 98.87%, the suggested technique surpassed all other methods. According to the study's findings, the suggested model can be used to further tasks like epilepsy diagnosis and recognising emotions by EEG classification. For signals like ECG and EMG, the suggested deep learning approach is also successful.

Non-linear techniques utilised for EEG signals are suggested by Sunil Kumar Prabhakar and Harikumar Rajaguru [4] as a systematic tool for the identification of alcoholism. The Correlation Dimension (CD) was used by the researchers for the first clustering and feature extraction. The feature extraction process was subsequently improved upon using several distance metrics, including correlation distance, city block distance, cosine distance, and chebyshev distance. Twelve classifiers were used by the researchers, including two newly suggested classifiers, the Adaboost.RT classifier and Modified Adaboost.RT classifier, as well as other classifiers for classification, including Random Forest, Artificial Neural Networks (ANN), Support Vector Machines (SVM), Naive Bayesian Classifiers (NBC), K-means classifiers, and the K Nearest Neighbour (KNN) Classifier. When CD and the suggested Modified Adaboost were used with correlation distance metrics, the results demonstrate an excellent accuracy in classification of roughly 98.99%.

The separable EEGNet (S-EEGNet), based on the Hilbert-Huang transform (HHT) and a separable convolutional neural network (CNN) with bilinear interpolation123, is a novel type of EEG classification network, according to the authors [5]. The study emphasises that while electroencephalogram (EEG) classification offers a wide range of potential uses, it is challenging for conventional neural networks to accurately capture the EEG signal's characteristics4. The EEG signal may be more accurately represented thanks to HHT, which transforms the EEG signal into a time-frequency model. In publication [6], the authors suggest an improved categorization and analysis of EEG data to aid in enhancing detection performance. The study emphasises the use of electroencephalography (EEG) as a diagnostic tool for neurological and mental problems. The authors suggest a novel method for improving EEG classification efficiency by making use of information about the physical location of EEG sensors. The authors suggest a novel method for improving EEG classification efficacy by making use of information about the physical location of EEG sensors. To enhance the categorization performance of EEG data, they simulate the spatial arrangement of the EEG sensors using a factorization-based technique.

A CNN-based training and categorization approach for EEG emotional characteristics is suggested in the study [7]. For the purpose of simultaneously learning spatial data as well as time-bound association via initial motor imagery EEG (MI-EEG) signals, the authors suggest combining a long and short-term storage network with spatial CNN. The report emphasises how active research in motor imagery EEG (MI-EEG), which can be used to identify a subject's active intent, has emerged recently. Since there aren't many meaningful elements in MI-EEG signals, reliable decoding is crucial for BCI devices to function properly but is also exceedingly difficult.

An automated deep learning-based method is suggested in the research paper [8] for extracting sleep apnea cycles from EEG information. The EEG data were divided into several modes using the variational mode decomposition (VMD) method in the study to extract effective processing of the fluctuations that occurred in the range of frequencies during apnea occurrences. The research suggests using a fully convolutional neural network (FCNN) to parallel extract the temporal information from each VMD mode while preserving their temporal relationships. An array of bi-directional long short term memory (LSTM) layers is used to jointly optimize the data collected from various EEG modes in order to examine the inter-modal temporal fluctuations. In order to improve classification performance, a semi-supervised strategy is investigated in this subject-neutral study. The suggested method's average accuracy during cross validation was 93.22%, 93.25%, and 89.41%. The proposed approach, according to the scientists, delivers notable performance gains in identifying sleep apnea occurrences and can be used in real-world settings.

A strategy for a hierarchical flow convolutional neural network (HF-CNN) model for reliable forearm movement categorization in accordance with elaborated rotation angles utilising electroencephalogram (EEG) information is described in paper [9]. The authors tested their method through trial and BNCI Horizon 2020 datasets to categorise forearm motions via EEG signals. The proposed approach showed a mediocre performance in terms of classification2. The authors of this paper concentrate on classifying forearm movements using EEG signals in accordance with elaborate rotation angles. For robust classification, they suggest using a HF-CNN model. BNCI Horizon 2020 and experimental datasets are used to evaluate the suggested technique.

A 3-D Convolutional Neural Network approach is suggested by the authors [10] for precisely detecting EEG abnormalities from multi-channel EEG inputs. The study builds a dataset 10 times larger than other datasets of this sort,

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preprocesses all channel EEG signals into data structures that may capture EEG spatio-temporal character, builds and trains a 28-layer deep residual network, automatically extracts characteristics from the initial information, and then categorises EEG signals. In conclusion, the research introduces a deep learning technique that can precisely identify EEG anomalies from multi-channel EEG inputs. The programme takes as input EEG signals from all channels and preprocesses them into data structures that can capture the spatio-temporal characteristics of the EEG signals. A 28-layer deep residual network is then built and trained to autonomously derive features from the initial data and categorise EEG signals.

III. Methodology

A. Dataset Description

The "EEG Dataset" consists of the EEG readings from 64 electrodes placed on the scalp of 122 individuals, both alcoholics and controls. Images of objects chosen out of the Snodgrass and Vanderwart picture set from 1980 were shown to each person as either one stimulus (S1) or two stimuli (S1 and S2). The S1 stimulus and the S2 stimulus were either identical (matched) or dissimilar (non-matched). For each person, there are 120 trials in the dataset, and each trial features a unique stimulus. The EEG recordings were recorded at 256 Hz for 1 second, and the electrode placements were situated at standard sites as specified by the American Electroencephalographic Association (AEA) in 1990. Every experiment is kept in a separate directory and has a header section with the meeting criteria and the subject identifier. The fourth letter of the subject identifier designates whether the individual was an alcoholic (a) or a control subject (c). The types of criteria that correspond include the single object is shown (S1 obj), the item is shown in a corresponding condition (S2 match), and the object is displayed in a non-matching condition (S2 no-match).

TABLE I. Acquired Data Sample

Trial #	Position of Sensor	Sample #	Measured Voltage (μV)
79	C1	6	-1.404
79	C1	7	1.038
79	C1	8	0.549
79	C1	9	-1.892
79	C1	10	-0.427

Table I shows a small sample of observed values of each of the 120 trials in the acquired EEG Data that is kept in its own file.

Each file contains data in the same format from various sensors. The trial number, sensor position, sample number (varying from 0 to 255), and sensor value in microvolts are all four fields of information. The number of trials, channels, and samples, as well as the sampling rate and time duration of each epoch, are also listed in the file header.

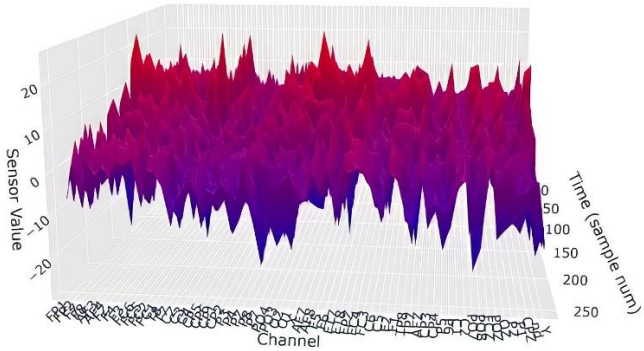

Fig. 1. 3D Heatmap of the sensor readings of each channel.

Figure 1 depicts the 3D visualization of the collected EEG data where the x and y axes correspond to the 64 input channels and the time at which values were sampled, along with the z-axis representing the voltage value recorded by the sensor.

Three distinct versions of the dataset are available: the little data set, which has information about just two subjects; the large data set, which includes information about twenty people; and the entire data set, which includes information about all 120 trials for all 122 subjects.

B. Implementation

In this study, we employed three machine learning algorithms alongside autoencoders to classify subjects as either alcoholic or non-alcoholic based on EEG data.

- Logistic Regression: A linear model that predicts the probability of the target class using a logistic function.
- Random Forest Classifier: It is an ensemble technique that combines the predictions of numerous decision trees that have been trained on various subsets of data.
- AdaBoost Classifier: An ensemble model that trains multiple weak classifiers sequentially, with each subsequent classifier focusing on the misclassified samples of the previous classifiers.

For logistic regression, we used the implementation provided by the scikit-learn library with default hyperparameters. The input features were the EEG measurements from the 64 electrodes, and the output was the binary classification of alcoholic or non-alcoholic.

For random forest classifier and AdaBoost classifier, we also used the implementations provided by scikit-learn with default hyperparameters. We set the number of estimators for both algorithms to 100, and the maximum depth for the random forest classifier to 10. As with logistic regression, the input features were the EEG measurements, and the output was the binary classification.

We evaluated the performance of each classifier using several metrics mentioned below:

- Accuracy: The percentage of forecasts that are accurate overall.

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- Recall: The percentage of actual positives (i.e., correctly predicted alcoholic subjects) out of all actual alcoholic subjects.
- Precision: The proportion of true positives out of all predicted alcoholic subjects.
- F1 Score: A weighted average of precision and recall, with a higher value indicating better performance.

The accuracies of the above mentioned models were not appreciable and hence to further improve these classification metrics, we explored the autoencoder architecture.

An auto-encoder is an unsupervised neural network which is used for representation learning. Generally, auto-encoders are used in applications like computer vision, natural language processing. They can also be used to work with large datasets which generally require more time to train or work with. In an unsupervised manner, the objective of an autoencoder is to acquire a compressed representation or encoding of the input data. This is generally done when the input data is large or has a high dimension. By compressing the data into a lower-dimensional space, auto-encoders can learn about the input data, or representations of the data more efficiently.

In mathematical formulae, auto-encoders are defined as a function:

$$f(x) = z \tag{1}$$

$$g(z) = x' \tag{2}$$

where x is the input data, z is the compressed data/representation and x' is the reconstructed output data. This is typically achieved through a series of layers that progressively reduce the dimensionality of the data.

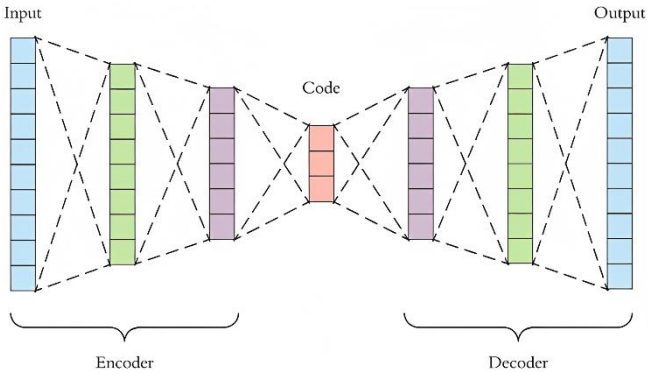

Fig. 2. Autoencoder architecture

Figure 2 shows the architecture of autoencoder which consists of encoder, bottleneck and decoder layers.

Initially, the input data is compressed by learning about the most important features of the dataset. Then, the compressed data is stored in the bottleneck of the network representation. The quantity of knowledge that may go throughout the entire network is constrained by a bottleneck (or the compressed code component), which forces a learnt compression of the provided input. In simple words, bottleneck forces an auto-encoder to learn a compressed representation of the input data. The decoder part in the figure is responsible for reconstructing the original input from the compressed representation generated by the encoder. Given the encoded representation z, the decoder applies a series of transformations to reconstruct the original input. More formally, the decoder function can be represented as:

$$x_hat = g(f(z)) \tag{3}$$

where f is the encoder function that maps the input x to the encoded representation z, and g is the decoder function that maps the encoded representation z back to the reconstructed input x_hat. In this research paper, it is aimed to represent a variation of auto-encoders where, the decoder part is not explicitly included and instead of reconstructing the input sequence, this architecture aims to capture the temporal dependencies and patterns in the data. Also, the purpose is to learn a compressed representation rather than explicitly reconstruct the original input.

Here, the autoencoders have been implemented using Keras, a popular machine learning framework. The auto-encoder model takes in EEG signals with shape (6,1), meaning each input consists of six scalar values. The model has several layers that work together to learn a compressed representation of the input.

- 64-unit dense layer with ReLU activation applied that linearly transforms input before activating the result nonlinearly.
- Two bidirectional LSTM layers, each with different numbers of units and dropout regularization applied to their outputs. Bidirectional LSTMs process input sequences in both directions, which can help capture dependencies in the input data more effectively.
- The output of the LSTM layers is flattened into a vector with a single dimension.
- With 128 units and a ReLU activation mechanism, an additional dense layer.
- Finally, a single sigmoid-activated dense layer produces the output of the auto-encoder, which is a single scalar value representing the compressed input.

The most crucial element employed in the aforementioned autoencoders is Long-Short Term Memory (LSTM). LSTMs are essential for sequential data. They learn long-term dependencies and handle high dimensions. LSTMs use output, forget, and input gates to control information flow. They have a cell state for retaining important data. LSTMs excel in processing long sequences by selectively remembering or forgetting information over time.

In this paper, LSTMs are used to analyze the information flow of the input sequence in either direction, from past to future or future to past. To improve accuracy, an upgraded variant called Bidirectional LSTMs is employed. These Bi-LSTMs process the input sequence in both forward and backward directions simultaneously, allowing the model to capture information from both past and future contexts. This dual processing enhances performance in sequential learning tasks, providing improved weight updation compared to vanilla LSTMs.

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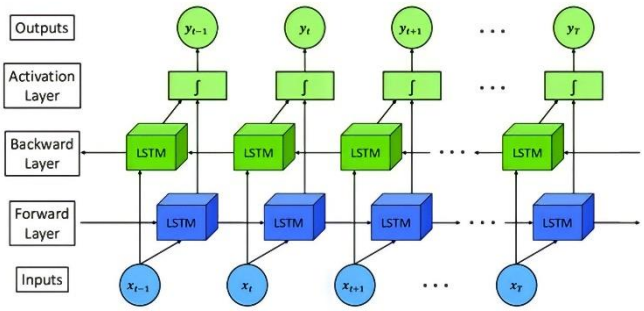

Fig. 3. Bidirectional LSTM architecture

Figure 3 shows the Bi-directional LSTM's architecture. Considering the current time-stamp or the current node (x_t), current input is sent in both the layers, forward (x_{t+1}) and backward (x_{t-1}) to analyze its future and previous values respectively.

Based on this data, whatever value is achieved, Rectified Linear Unit (ReLU) activation function is applied (except the final layer which consists of Sigmoid activation function) to achieve an output. For the next neuron, the same process will take place. ReLU will not only help in the faster execution of the autoencoder but will also solve the 'dead-neuron' problem which could be a potential hazard when working with (temporal) EEG data which consists of important features such as time stamp, sensor value, etc.

The input data includes electrode positions on the scalp, representing different channels of data. The bi-directional LSTM model captures relationships between these channels and their response to various stimuli. It takes EEG recordings from all 64 electrodes as input, learning temporal dependencies between them. The algorithm utilizes a Bi-LSTM to build connections in both forward and backward directions, incorporating information from the recent past and future.

Model: "model"

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, 6, 1)]	0
dense (Dense)	(None, 6, 64)	128
bidirectional (Bidirectional	(None, 6, 512)	657408
dropout (Dropout)	(None, 6, 512)	0
bidirectional_1 (Bidirection	(None, 6, 256)	656384
dropout_1 (Dropout)	(None, 6, 256)	0
flatten (Flatten)	(None, 1536)	0
dense_1 (Dense)	(None, 128)	196736
dense_2 (Dense)	(None, 1)	129

Total params: 1,510,785
Trainable params: 1,510,785
Non-trainable params: 0

Fig. 4. Model summary of Autoencoder model

Figure 4 depicts the model summary of the employed auto-encoder model.

In this specific autoencoder architecture, the input layer is defined with a shape of (6,1), indicating that it expects input data with six features consisting of trial number, sample number, sensor value, matching condition, channel and time (the independent features) and one-dimensional data. Two 'hidden' dense layers with units to be 64 and 128 are used to connect all the neurons in the hidden layers to the previous layers. In addition to this, ReLU activation function is used in both the layers and specifically in the first dense layer, regularization in the form of L2 regularization is applied to this layer to prevent overfitting. Bi-LSTMs are stacked with 256 and 126 units which are the main backbone of the autoencoder model responsible for forward and backward travel in order to get an output for current neuron. This also allows the network to learn more complex temporal patterns in the input sequence, as each LSTM layer can learn different features of the sequence which can identify complex patterns. Dropout layers are, sequentially, added to every Bi-LSTM layer in order to prevent overfitting, make the model more robust and help model generalize over unseen data in a much more efficient way. Finally, sigmoid activation function is applied to the output layer 'binary-classify' in terms of alcoholism.

To augment this prediction, hyperparameters are used which will help in more robust and assurance in the prediction. The hyperparameters used in the autoencoders are as follows:

- **Adam Optimizer** - During training, the autoencoder function uses Adam optimizer to reduce loss. It is a widely used stochastic gradient descent optimization algorithm, suitable for deep neural networks. The optimizer adjusts the learning rate, which is assigned 0.001 based on empirical evaluation. This value can affect the training process's speed and quality. The Adam optimizer is utilized in both the autoencoder function and the classification algorithm of the "EEG Correlates of Genetic Predisposition to Alcoholism Dataset" to optimize the weights and parameters for classifying alcoholic and non-alcoholic subjects.

- **Early Stopping** - EarlyStopping stops the training process when the validation loss has not improved for a specified number of epochs, as determined by the "patience" parameter which is set to 10 in our case. The "monitor" parameter is set to "val_loss", to monitor validation loss during training. The "mode" parameter is set to "min", indicating that the When the monitored quantity (in this instance, the validation loss) stops dropping, EarlyStopping will halt the training. When the amount decreased, it would halt if the "mode" had been set to "max".

- **Learning Rate Scheduler** - The Learning Rate Scheduler adjusts the learning rate of the optimizer during training, based on the current epoch, as determined by the provided lambda subroutine. 0.001 is set as the beginning learning rate and after a period of ten epochs it decreases progressively by a multiple of 10. These techniques can help prevent overfitting and improve the accuracy and stability of the autoencoder function.

During the training of the autoencoder, the Binary Cross-Entropy Loss function is employed to assess the loss in each epoch. This particular loss function is frequently utilized to quantify the disparity between the predicted probability

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distributions and the actual distributions. It is commonly used in deep learning models for binary classification problems. The binary cross entropy loss function is passed as a parameter to the 'loss' argument in the model compilation step. The model's efficacy throughout training is also evaluated using the 'accuracy'. This statistic counts how many subjects in the dataset were accurately categorised out of all the cases.

We observed that with a batch size equal to 512 and epoch set to 5, the loss, from epoch 1 through epoch 5, reduced from 0.6378 to 0.5960.

IV. Results

Our proposed auto-encoder model achieved robust and accurate prediction performance, even in the presence of noise and other causes of uncertainty in the EEG data. Several metrics, including accuracy, precision, recall, and F1 score, were utilized to evaluate the model's effectiveness. On the test set, the model achieved an accuracy of 58.4% with AdaBoost Classifier, 58% with Random Forest Classifier, 55% with Logistic Regression, and finally, 65.5% with AutoEncoders, highlighting its efficacy in predicting alcoholism from EEG signals.

The comparison between the given model architecture (based on Bidirectional LSTMs) and other models like Random Forest, Logistic Regression, and AdaBoost depends on various factors and the specific characteristics of the EEG dataset. Few aspects that are inferred from this are as follows:

1) Data Representation: The model architecture based on Bidirectional LSTMs is particularly well-suited for equential data, such as time series or sequential patterns in EEG signals. LSTMs can effectively capture long-term dependencies and temporal dynamics in the data, which might be crucial in EEG signal analysis. On the other hand, models like Random Forest, Logistic Regression, and AdaBoost are typically better suited for tabular or structured data where the order of the data points does not matter.

2) Feature Learning: Bidirectional LSTMs have the ability to automatically learn meaningful features from the input data as part of the training process. This can be advantageous in cases where the explicit definition of features is challenging or when the relevant features are not known in advance. In contrast, models like Random Forest and Logistic Regression require handcrafted features, and the quality of their performance heavily depends on the choice and engineering of relevant features.

3) Handling Non-linearity: Bidirectional LSTMs are known for their ability to model complex non-linear relationships in the data. This flexibility can be advantageous when dealing with EEG signals, as they often exhibit non-linear patterns and dynamics. In contrast, models like Random Forest and AdaBoost can also capture non-linear relationships to some extent, but they might not be as powerful in capturing complex non-linear patterns as deep learning models.

TABLE II. Metrics Comparison

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	55.7	57	45.3	50.5
Random Forest	58	59	51.5	55
AdaBoost	58.4	61.1	45.7	52.3
AutoEncoder	65.5	70.2	47.6	56.7

Table II compares the common performance metrics obtained on the EEG test dataset between all the utilized algorithms in this study. It is observed that the accuracy score of AutoEncoder model surpasses the others.

The 2 metrics observed during training of the AutoEncoder model over 5 epochs were accuracy and loss. The change of the metrics with each epoch is depicted in the following diagrams.

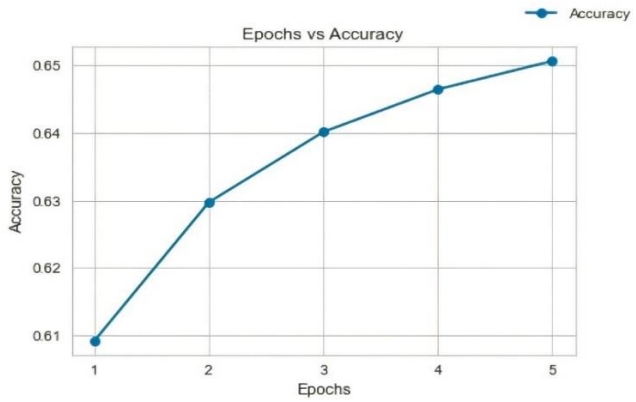

Fig. 5. Epochs vs accuracy of Autoencoder

Fig. 5. represents the observed change in accuracy with each of the 5 epochs during the training process. It can be seen that the accuracy increases with every epoch from 0.61 to 0.65 which is higher than other algorithms.

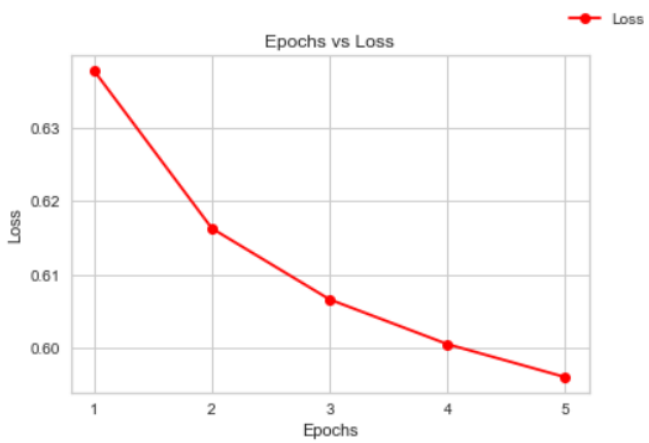

Fig. 6. Epochs vs loss of Autoencoder

Fig. 6. similarly shows the change in loss of the model at the end of each training loop which takes input batch size of

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512. During each epoch, after training on a set amount of data, the model loss is evaluated during validation cycle and is observed to decrease continually.

V. Conclusion and Discussion

In summary, the auto-encoder model suggested in this study exhibited notably better predictive performance than conventional machine learning algorithms in predicting alcoholism from EEG signals. This finding suggests that auto-encoders hold promise as an effective approach for extracting meaningful features from complex and noisy biomedical signals, with potential applications in the diagnosis and prediction of other types of biomedical data. Few of the research papers share the common goal of utilizing deep learning techniques to improve the accuracy of EEG signal analysis for various purposes. Each paper focuses on different applications and approaches, including 1D and 2D representations of EEG channels and the variational mode decomposition (VMD) algorithm. Paper [2] introduces a novel 2D-IDR model that considers the spatial configuration of EEG sensors to improve classification accuracy. Results show consistent 5%-8% higher accuracy compared to the one-dimensional model, regardless of features or classifier type. The proposed autoencoder model handles complex and noisy EEG data, learning meaningful representations and improving classification accuracy. Autoencoders offer flexibility for various biomedical signals, making them suitable for diverse applications. Overall, autoencoder models show promise for EEG-based classification tasks, outperforming traditional approaches like the 2D-IDR model.

The proposed model in [8] uses a fully convolutional neural network (FCNN) hinder with causal dilated convolutions to concurrently extract spatial characteristics from every VMD modes in a parallel behaviour to identify sleep apnea frames in EEG signals while maintaining the temporal dependencies between each VMD mode. In contrast, this research paper introduces an auto-encoder architecture for predicting alcoholism from EEG signals. Although both models employ deep learning techniques, the model designed for detecting sleep apnea in EEG signals has a more intricate architecture compared to the auto-encoder model, as it incorporates dilated convolutions and parallel processing of VMD modes to extract features, while the auto-encoder architecture is relatively simpler.

Du Yun-Mei et al. [10] used a 3D CNN to detect abnormal EEG signals from multi-channel EEG data. They created a dataset with 14049 samples from different datasets and used a 28-layer deep residual network to extract features and identify anomalies. The paper emphasizes the importance of using 3D frame data and deep CNN models for EEG classification. Our research and [10] focus on different aspects. [10] focuses on accurate detection of abnormal EEG signals using 3D CNNs and a large annotated dataset, while our paper focuses on using an auto-encoder model to predict alcoholism from EEG signals. The studies differ in complexity, with [10] using more complex neural network architectures and larger datasets.

VI. Future Scope

In future research, exploring complex neural network designs like convolutional neural networks can improve the accuracy of auto-encoder models when dealing with different types of EEG data. Techniques such as feature engineering and data preprocessing can help extract useful information from EEG signals, leading to better performance of these models.

A critical aspect in gaining a comprehensive understanding of the limitations and generalization capabilities of auto-encoder models lies in the collection of larger and diverse datasets. By including individuals with varying levels of alcoholism severity and those with co-existing mental health conditions, we can achieve more accurate classifications. By analyzing EEG signals during tasks related to cognitive control and inhibition, we can detect changes in brain activity that can help with the diagnosis and understanding of alcohol use disorder.

In addition to Deep Learning and Computer Vision techniques, there are other methods that can benefit alcoholism research. For example, Natural Language Processing can provide valuable insights from patient interviews and reports, giving us a better understanding of alcoholism symptoms and behaviors. Similarly, speech recognition can analyze spoken language to identify patterns and features related to alcoholism. Exploring these alternative approaches allows us to collect a wider range of data and improve the diagnosis and prediction of various health conditions, including alcoholism.

The following suggestions can be incorporated in the future review of this research paper:

1. Increasing the capacity of the model by adding more layers, increasing the number of units in the existing layers along with different activation functions. This can help the model learn more intricate patterns and representations from the EEG signals.

2. Fine tuning existing hyperparameters such as adjusting the learning rate, dropout rate or the regularization strength can significantly impact model convergence and accuracy. Adopting new learning rate schedules or using adaptive learning rate techniques such as Cyclical Learning Rates or ReduceLROnPlateau can also improve performance.

3. Applying data augmentation techniques specific to EEG signals such as time shifting, scaling, and adding noise can artificially increase the size of your dataset and improve the model's ability to generalize.

4. More advanced architectures such as variational autoencoders (VAEs) or convolutional autoencoders (CAEs) that are specifically designed for capturing spatial and temporal patterns in the data and may provide better accuracy for EEG signal analysis.

5. Regularization techniques can help prevent overfitting and improve generalization. Apart from the L2 regularization applied to the current model, we can explore other regularization techniques such as dropout or batch normalization to different layers of the model.

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